Mohammed Mohi Uddin

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Signal processing and efficient ML under constraint

Final year computer science student working on systems where **the failure mode matters more than the headline metric**. Available full time for twelve or more weeks in **summer 2027**, willing to relocate.

Research interests

Signal recovery under motion and noise, and trust gates that discard corrupted measurements rather than smoothing over them. **Cost efficient inference**: when a learned component earns its compute against a deterministic baseline. **Uncertainty and provenance** in generated or reconstructed outputs.

Education and recognition

B.Tech, Computer Science Engineering (Data Science) Aug 2023 – Aug 2027

Swami Vivekananda Institute of Technology, Hyderabad · CGPA 7.85 / 10 · top 25% of 120 students

- OptiXon design reviewed with **M.Tech students at IIT Hyderabad** and a working hardware professional, who confirmed feasibility and suggested changes. Every design decision is my own.
- Smart India Hackathon 2025, rural telemedicine PS25018. IIT Hyderabad certificates in Generative AI with ML and Data Science.

Research projects

OptiXon · photoplethysmography in prescription eyewear 2024 – Jul 2026

Founder and technical lead · Second place, Techno Udbhav 2025 research competition

- Continuous heart rate from the superficial temporal artery through a sensor in an eyeglass temple, where **motion artifact, not sensor selection, is the dominant problem**. The IMU acts as a trust gate: the gyroscope flags a corrupted window before processing, so bad samples are discarded rather than averaged into a confident wrong reading.
- Verified the lithium protection circuit at **three independent levels** after finding a pin mapping error that would have defeated all protection while passing every functional test. Battery and thermal figures are design estimates and labelled as such, since no prototype exists.

`nRF52832, MAX30102 PPG, LSM6DSOX IMU, KiCad, ngspice, C`

Multimodal classification under a small label budget 2024, rebuilt 2026

- Fused a convolutional image embedding with a sparse TF-IDF representation of OCR text under a linear head: **96.92% across 20 classes** on a 260 image held-out set, from 1,297 labelled examples. The gain came from **error analysis rather than capacity**, and the linear head kept each modality inspectable.

Deterministic matching with selective learned fallback May 2025 – now

- Where a learned matcher earns its cost against a rule based one, tested on transaction reconciliation because ground truth is constructible. **Blocking reduced a 9.8 billion pair comparison** to a tractable candidate set: 95% matched, zero mismatches against a synthetic answer key, 4.25 seconds at 100,000 records, zero inference calls.

Retrieval augmented generation · IIT Hyderabad summer programme Jun – Jul 2025

- Found **chunk boundary selection, not prompt design, dominated answer quality**, and that retrieval failure produced fluent confident wrong answers with no distinguishing signal. That observation drove later work on explicit verification state in a deployed system.

Technical skills

Programming: Python (primary), TypeScript, C, SQL

ML and signals: PyTorch, TensorFlow, scikit-learn, OpenCV, embeddings and retrieval, evaluation design, PPG and IMU processing, motion artifact rejection, ngspice, KiCad, nRF52

Mathematics: Linear algebra, probability and statistics, optimisation, numerical methods.